

AI-Based Early Detection of Diabetes Complications: Balancing Accuracy and Interpretability Across Multi-Source Repositories

Abstract

Diabetes is a serious health condition specially prominent in a country as vast as India. Around 40-50% of diabetics are undiagnosed. Around 1 in 4 diabetics (25%) are left undiagnosed in some population studies, India currently has one of the highest numbers of diabetics in the world.

Around 10.5% of adults (20-79 years) have diabetes. If diabetes isn't found early, it can cause serious consequences. According to previous data diabetes now causes about 3.1% of all deaths in India. Late diagnosis as a major indirect cause of the deaths including poor management.

Old computational system do prediction on single dataset (PIMA) and limited features (Volatile glucose) that doesn't provide real clinical trust. To overcome that problem we proposed hybrid machine learning framework that evaluates to different datasets (PIMA and Medical City Hospital Data).

We found that the accuracy was low on simple datasets, but on datasets with advanced medical parameters, the XGBoost ensemble achieved the highest accuracy of 98.3%. We used SHAP to show that the model gives higher priority to long-term markers (HbA1c and cholesterol), which will help doctors build trust in the model.

Keywords: Diabetes Mellitus, Machine Learning, XGBoost Ensemble, Explainable AI (XAI), SHAP Framework, Multi-Dataset Validation.

1. Introduction

- **Clinical and Global Context**

The majority of people worldwide suffer from diabetes, which is one of the most prevalent illnesses. Additionally, this illness puts a strain on hospital employees and doctors. If diabetes is not detected early, it can lead to major problems such as diabetic retinopathy, nephropathy, neuropathy, and cardiovascular illnesses, which shorten our lives and cause disability. International health reports state that diabetes affects a significant number of people. Urine testing was utilised for detection in ancient Egypt, Greece, and India. As medical science has advanced, more sophisticated techniques, including the HbA1c test and blood glucose test, are now used. In order to reduce long-term health concerns and enhance patient outcomes, prompt detection of these problems is crucial. This issue is becoming a major concern in a large country like India, which is currently among the countries most affected by diabetes worldwide.

Around 10.5% of adults in India are living with diabetes. What is even more worrying is that nearly 40-50% of cases are diagnosed very late, making timely treatment difficult. As a result, diabetes is responsible for 3.1% of deaths in the country.

- **Role of AI and Rural Healthcare Challenges**

Artificial intelligence and machine learning techniques have made it simple to promptly diagnose people who are at risk. AI helps with treatment decision-making in addition to being essential for prediction and automated diagnosis. AI systems can analyze patient data and aid in the development of novel medications. It also recommends ideal fitness regimens and nutrition charts. Additionally, it keeps track of the patient's everyday health. AI also recommends a suitable meal chart, which makes it easier to manage a diabetic diet by calculating the number of calories, carbs, and sugar in food. In addition, AI can identify a patient's mental wellness. AI is also improving its ability to monitor without discomfort, and a new AI system can reveal the cause of diabetes. In a step-by-step structure, AI analyzes a patient's blood pressure, cholesterol, blood glucose, and BMI. Doctors have certain challenges when analyzing vast amounts of data, but AI makes it much simpler. AI is also capable of forecasting future medical conditions. There are several difficulties as well, such as data privacy concerns. AI is really

beneficial and makes our work easier, but it still requires high-quality data and the doctor's advice. Because there are fewer medical professionals and facilities in rural locations, AI remote diagnosis can be highly helpful. We can also set up an inexpensive AI screening camp. Additionally, this saves people's time and money and is especially beneficial for those in poverty who are unable to spend much money because of their dire financial circumstances.

- **The Performance-Interpretability Trade-off (The Research Gap)**

In order to ensure both performance and clinical dependability, this research focuses on developing an AI-based model for the early diagnosis of diabetic complications while striking a compromise between model accuracy and interpretability. AI can help doctors make decisions, but it cannot take their place. Although diagnosis is improved, there are still issues that could be resolved in the future. The usage of AI is growing faster over time and has become a crucial component of healthcare; the sooner diseases are identified, the better human life will be. More information can be gathered to improve healthcare planning. It shows that AI research is dependable, effective, sophisticated, and useful for overcoming obstacles. For practical healthcare applications, it is crucial to provide both high performance and clinical reliability. In this paper, we test our ideas on two different Data sources: the PIMA dataset and the Medical City Hospital dataset. We also use the SHAP tool to explain the computer's decisions so that doctors can easily understand and trust the results.

2. Literature Review

• Machine Learning Applications in Diabetes Prediction

Nowadays lots of Patient's data was collected in the hospitals that's why researcher used Machine learning algorithms to detect diabetes. It saves the doctor's time and also reduced the chances of mistakes. Previous researchers like **Sisodia & Sisodia (2018)** and **Naz & Ahuja (2020)** used **Machine learning and Deep learning** methods for the prediction. However their work depended on the American PIMA Indian dataset, mainly because access to multisystem clinical marker data was limited at that time. The present study fulfills this research gap by performing a comparative evaluation using both the Pima dataset and real clinical data collected from Medical City Hospital.

As the time passed, researchers shifted to focus on tracking of severe complications (organ damage) instead of just predicting diabetes. Researchers like **Gulshan et al. (2016)** were successful in detecting diabetic retinopathy from the retinal photos using deep learning methods, However these advanced models were complex and also created "**Black Box**" issue that's why **Tasin et al. (2022)** suggest Explainable AI to build trust in healthcare, In the past studies Explainable AI tools (like SHAP) were mentioned but no one checked that when data is switched from simple to advanced hospital registry how features preference changes.

Table 1: Comparative Analysis of Prior Literature and Proposed Framework

Research Reference	Core Algorithm Paradigm	Evaluation Dataset Platform	Clinically Meaningful XAI?	Primary Research Bottleneck / Gap
Sisodia et al. (2018)	Standard Machine Learning	PIMA Repository Only	No	Restricted feature dimensions; focused only on raw metric maximization.
Gulshan et al. (2016)	Advanced Deep Learning	Localized Retinal Images	No	Narrow diagnostic scope limited strictly to single complication tracking (Retinopathy).
Tasin et al. (2022)	ML+ Explainable AI	Legacy Data Sources	Yes (SHAP Enabled)	Lacked cross-dataset validation on advanced multi-system patient biomarkers.
Our Proposed Study	Hybrid Ensemble Optimization	PIMA+ Medical City Hospital	Yes (SHAP Enabled)	Successfully resolves the performance-interpretability trade-off across distinct clinical profiles.

3. Proposed Methodology

3.1 Data Pipeline and Processing

In the first step of our Methodology we organized and cleaned the raw columns of both datasets (PIMA and Medical City). Since the Medical City dataset contains text columns like (Gender and Class), we used a **Label encoder** to convert these text fields into numbers so that computer algorithm could easily understand them. We used the **Median value** to fill up the blank cells in the data to avoid errors during model training. At the end to scale features properly, we used **Standard Scaler** and split the data matrix into 80% training and 20% testing.

3.2 Model Selection and Hybrid Architecture

To properly analyze diabetes classification, we developed a hybrid architecture in which we used the classification capabilities of four different algorithms. We selected Logistic Regression and Decision Tree for the interpretable models because they are simple and can be easily accepted in medical systems, as their internal decision-making process is quite easy to understand and explain. To handle complex patient features and maximize prediction accuracy, we also used two powerful tree-based ensemble methods: Random Forest and XGBoost classifiers. Our main objective is not just to achieve a high accuracy score, but to compare the performance behavior of all four models across different data environments, such as PIMA and Medical City datasets.

3.3 Explainable AI (XAI) Framework

Although advanced models like XGBoost and Random Forest often provide very high accuracy, they are usually considered “black box” models because doctors cannot clearly understand how the system reaches its decisions internally. Due to this lack of transparency, medical professionals may hesitate to trust such models blindly. To address this issue, we integrated the SHAP framework with our best-performing model. SHAP calculates a mathematical impact score for each patient feature, such as HbA1c, glucose level, and BMI, making it easier to identify which factors are increasing the risk of diabetes. This feature ranking also helps us verify whether the model’s decision-making aligns with real medical science and clinical understanding.

4. Experimental Results and Discussion

4.1 Comparative Accuracy Matrix

Table 2: Computed Classification Accuracy Across Distinct Repositories

Target Dataset Platform	Logistic Regression (Baseline)	Decision Tree Classifier	Random Forest Ensemble	Advanced XGBoost Ensemble	Top Explanatory Marker (SHAP)
PIMA Indian Dataset	75.32%	74.68%	72.08%	72.08%	Glucose Concentration
Medical City Dataset	94.00%	96.50%	99.00%	98.50%	HbA1c & Cholesterol

When we look at Table 2, all the models perform almost similarly on the PIMA dataset, with accuracy between 72% and 75%. Among these, Logistic Regression gave the highest baseline accuracy of 75.32%. After switching to the Medical City dataset, the accuracy of all models increased significantly. Random Forest achieved the highest accuracy of 99.00%, followed by XGBoost with 98.50%. These results show that when models are trained on more detailed clinical data, advanced tree-based models perform much better than simple linear models.

4.2 Post-Hoc Explainability and Feature Attribution

Only getting high accuracy number is not enough for clinical validation, we used SHAP framework, so that we can understand and explain the internal decision patterns of the algorithms. On the PIMA dataset, the SHAP metrics clearly showed that glucose was the most important feature because advanced medical tests were not available in that dataset. when we moved to the advanced Medical City dataset, the model's computational behaviour changed completely. Instead of focusing mainly on fluctuating glucose levels, the model started giving higher importance to long-term health indicators such as HbA1c and cholesterol. This selection by the model matches real medical science very well, because doctors also rely more on stable three-month HbA1c profiles than normal fluctuating blood sugar levels when predicting complications and long-term health risks.

5. Limitations and Future Scope

The limitation of our current study is that we used a saved dataset which does not allow for real-time patient tracking. Our current dataset is limited to only a few thousand patients, which is still a relatively small sample size for a vast population like India. Both the PIMA and Medical City datasets still lack some very important medical parameters, such as the patient's exact diet chart, family history, genetic background, smoking habits, and physical exercise routine. Without these factors, the model cannot predict a completely accurate health profile. Another limitation is that our advanced dataset comes from only one specific medical facility. Since the data belongs to a single hospital, there is a possibility that the model may not provide the same level of accuracy for patients from different cities or countries. We have mainly tested advanced machine learning algorithms like XGBoost and Random Forest. However, we have not yet explored deep learning or neural network models, which may perform even better on larger datasets. Although SHAP helps us identify that HbA1c is the most important feature at the overall hospital level, it does not automatically generate a real-time local explanation for each individual patient on the diagnosis screen, which is known as local interpretability.

Our future plan is to connect this predictive engine with real hospital networks and we will also develop a web application that doctors can use and run in the real time. In the future, we plan to integrate additional biometric and clinical features, including kidney damage and heart-related health indicators, to make the model more advanced and capable of handling complex medical patterns. We also aim to design a dynamic Streamlit-based web dashboard that doctors can use in real time for quick analysis and prediction. In addition, the software model will be deployed on a lightweight cloud architecture so that testing and remote diagnostics can also be performed through mobile phones in rural health screening camps.

6. Conclusion

- **Multi-Repository Validation:** This study successfully validated diabetes risk screening by performing a comparative evaluation on two completely different datasets, PIMA and Medical City.
- **Ensemble Performance Peak:** When the machine learning models were provided with advanced laboratory features, the tree-based ensemble frameworks achieved the highest accuracy, where Random Forest reached 99.00% and XGBoost achieved a peak score of 98.50%.
- **Clinical Explanation Alignment:** By applying the SHAP framework, we were able to explain the model's black-box decisions and showed that the system was automatically giving top priority to important medical factors such as HbA1c and cholesterol, in line with real clinical guidelines.
- **Healthcare Scalability:** This safe and interpretable approach can help increase trust in clinical diagnostics among medical professions and may also become a reliable support system for low-cost diabetes screening in rural areas.